Driver scheduling for Ride-Hailing service

Group 23

1 Project overview

The primary operational challenge of this project is balancing driver supply with passenger demand. To do this, two main goals must be achieved: minimising passenger waiting times while simultaneously reducing the time drivers spend without a passenger.

Since our company lacks internal historical data, we used data from publicly available New York City Taxi records from 2024. This data set provided over 41 million records, offering a comprehensive view of how passengers currently move throughout the city. The raw data was transformed into a refined data set focused solely on the Manhattan market, using official taxi zone lookups to isolate 69 specific zones within Manhattan. Using this data, we have engineered a predictive framework that transforms raw historical patterns into a real-world simulation of the Manhattan market. This report details our findings and provides actionable insights for:

- **Demand forecasting:** Predicting how many rides will be requested every hour.
- **Revenue optimisation:** Identifying the most profitable times and locations to maximise driver earnings.

Ultimately, this report outlines a data-driven strategy for driver deployment that optimises both customer satisfaction and driver retention.

2 Data Preprocessing

To ensure that our predictions for demand and revenue are reliable, we implemented a rigorous multi-stage cleaning. Approximately 15% of the total records were removed due to data anomalies, such as negative fare amounts, zero distance, or missing vendor metadata. These records represent system errors and would have skewed our predictions. This also included records where the drop-off occurred before the pickup, ensuring that the data is chronologically sound. Extreme outliers, including trips longer than 5 hours or fares exceeding $100, were also removed to prevent them from distorting typical expectations.

Beyond cleaning, the models were taught to recognise patterns specific to New York City by engineering several key features. These included:

- **Lag features:** These look back at the previous 1-hour, 2-hour, and 24-hour periods to help the model predict immediate trends. If demand was rising at 4:00 PM, the model uses that momentum to project demand at 5:00 PM.
- **Time features:** Trips were categorised by hour of the day, day of the week, month, and season to account for daily and seasonal fluctuations.
- **Operational features:** Rush hour (7–9 AM and 4–7 PM) and weekends were explicitly tagged to help models differentiate between commuting patterns and leisure demand.

Finally, we shifted the focus from individual trips to Hourly Zone-Level Performance. We generated a gapless hourly timeline for every Manhattan zone, ensuring that even quiet hours, with zero recorded trips, were accounted for in our training. This processed data allows us to predict two critical targets simultaneously:

- **Ride demand:** To optimise driver deployment and minimise wait times.
- **Expected revenue:** To identify the most profitable times and locations for drivers.

3 Modelling Approach

To ensure the highest accuracy possible, we developed and tested six distinct machine learning models, three dedicated to predicting passenger demand and three to forecasting driver revenue. These models included:

- **Linear Regression:** Used as a baseline to capture general trends and linear relationships between time and demand.

- **Random Forest:** Exceptionally robust against outliers and can capture complex, non-linear relationships.
- **Neural Network:** Implemented to identify deep, complex dependencies in the data that traditional models might overlook.
- **XGboost (Extreme Gradient Boosting):** Selected for its speed and its ability to learn from previous errors, making it highly sensitive to subtle market shifts.

Although all models performed competently, we evaluated them primarily based on Mean Absolute Error (MAE). This provides the most practical metric, telling us on average exactly how many rides or dollars off our prediction is likely to be. While MAE tells us the size of our typical error, we also used R-squared (R^2) to measure the model's overall reliability. This represents the percentage of real-world fluctuations that our model successfully explains. We prioritised the models that offered the lowest average error and the highest stability (lowest standard deviation) across different times of the week to ensure reliable scheduling.

Based on our evaluation, we selected the following models for deployment:

- **Demand Prediction:** The Random Forest was selected due to its stability, highest R-squared (0.91), lowest MAE (13.1 rides) and clear feature importance rankings.
- **Revenue Prediction:** The XGBoost model was chosen as it provided the best balance of speed and interpretability. It yielded a high R-squared (0.90), retained a low MAE ($342), and also provided clear feature importance rankings.

4 Demand Analysis

Our analysis reveals that Manhattan ride demand follows a highly predictable rhythm. Our final prediction model (Random Forest) achieved an R-squared of 0.91, indicating that it successfully explains 91% of the variance in hourly demand. The Mean Absolute Error (MAE) of approximately 13 means that in any given hour, the staffing predictions are typically accurate to within 13 rides, a level of precision that allows for efficient driver scheduling without significant idle time.

Through feature importance analysis, we identified the primary drivers that dictate Manhattan demand:

1. **Historical Momentum (Lags):** The strongest predictor of demand is the volume recorded in the previous 60 minutes. Demand in busy zones stays busy.
2. **Temporal Features:** The specific hour of the day and whether it is a weekday or weekend significantly shift rider behaviour. This temporal dependence is illustrated in the heat map below (Fig.2).

It is apparent from the heat map that demand peaks during the early evening weekday work rush (16:00–19:00). This is also demonstrated in Table.1, which shows the peak demand and revenue values. We can also observe how demand starts later in the day on weekends compared to weekdays, providing clear indicators for optimal staffing schedules. Sunday exhibits the lowest overall demand, reflecting a transition from commercial activity to leisure and rest. This provides a strategic window for scheduled vehicle maintenance or driver downtime without impacting service. Although the average demand is predictable, we identified a critical relationship between volume and stability. As demand increases during the evening peak (18:00), the standard deviation of that demand also rises to its highest point (101.3). This means that while 6:00 PM is the busiest hour, it is also the hour where demand is most likely to fluctuate due to external factors like weather or traffic.

The bar chart shown in Fig.3 highlights the specific zones with the highest predicted volume, notably zones 237, 161, and 236. Identifying these zones is essential for driver planning, as they represent the most reliable locations for consistent trip turnover.

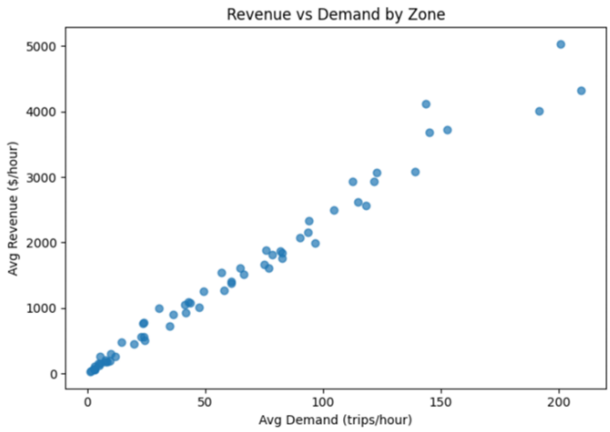

Figure 1: Scatter plot showing the positive linear relationship between Revenue and Demand.

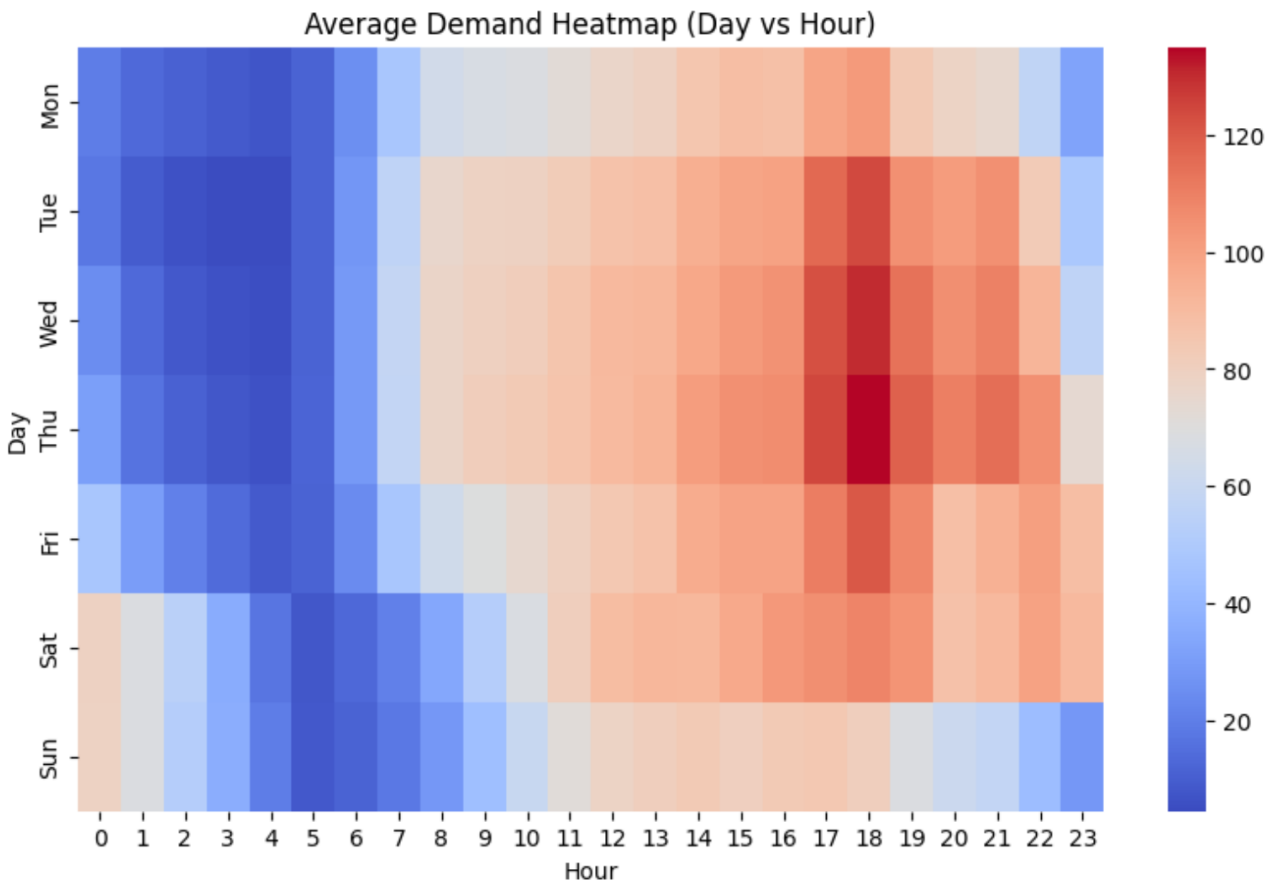

Figure 2: Hourly demand heat map, illustrating the concentration of ride requests by hour and day of the week. Dark red regions highlight high-volume of demand, while the dark blue regions indicate minimal demand.

Pickup hour	Demand (trips)	Revenue ($)
16:00	94	2342
17:00	100	2511
18:00	106	2550
19:00	94	2201

Table 1: Peak predicted Demand and Revenue by Hour (Manhattan Average)

5 Revenue Insights

While demand tells us where the passengers are, our revenue model identifies where the most profitable opportunities lie. Accurate revenue forecasting is important not only for corporate financial planning but also for ensuring driver earnings remain sustainable. Our final prediction model (XGBoost) achieved an R-squared of 0.90, effectively explaining 90% of the variation in hourly revenue across Manhattan. In practical terms, the model predicts the total hourly revenue with a typical error of roughly $343. Considering the high volume of Manhattan transactions, this provides a highly reliable baseline for financial planning.

Through feature importance analysis, we confirmed that the primary drivers of revenue mirror those of demand: historical momentum and temporal features. Our analysis indicates that revenue is fundamentally driven by trip density rather than high fare trips. As illustrated in Fig.1, there is a near-perfect positive linear relationship between ride volume (demand) and total earnings (revenue). Consequently, the revenue heat map also mirrored the demand patterns, again highlighting a power window between 16:00 and 19:00. The 18:00 (6 PM) window represents the highest earning potential, with an average hourly revenue of approximately $2,550 per zone. This peak starts to climb at 16:00 ($2,342), peaking at 18:00, and remaining high until 19:00 ($2,201). This suggests that the three-hour block from 16:00 to 19:00 is the most critical window for maximising profitability.

The top revenue-generating zones largely overlap with high-demand areas (161 and 237) as shown in Fig.3. However, the ordering of these

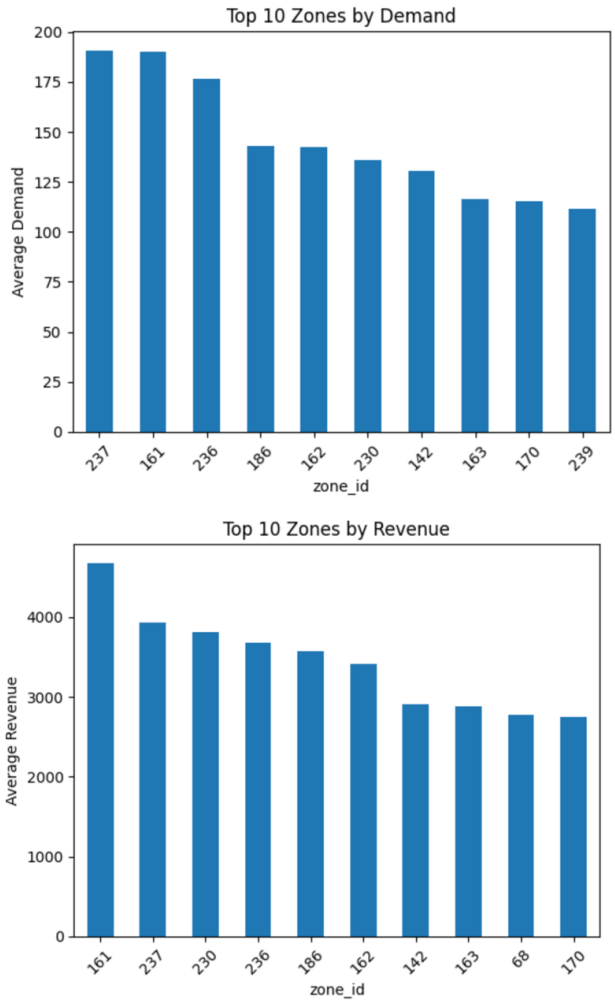

Figure 3: Top Manhattan zones by predicted demand (top) and predicted revenue (bottom).

zones vary slightly. This disparity suggests that certain neighbourhoods may yield higher revenue through longer-distance trips which increase the average fare per pickup.

6 Recommendations

Based on our predictive models, we recommend a data-driven staffing strategy for maximising driver yield and reducing passenger waiting times. We propose the following operational policy for Manhattan:

- **The late afternoon rush:** The peak window between 16:00 and 19:00 represents the most significant revenue opportunity, with total hourly earnings peaking at approximately $2,549. We recommend maintaining high driver density during these hours to capture this volume, reducing passenger wait times and maximising revenue.
- **Demand buffer:** Since the 18:00 peak demand exhibited high variability, a driver surplus during this time is recommended to account for unpredictable demand spikes and ensure service consistency.
- **Shift delay strategy:** Analysis indicated that weekend demand picks up significantly later than weekday demand. We recommend shifting the primary driver start times from 07:00 (workday standard) to 10:00 on Saturdays and Sundays. This ensures maximum driver availability during the higher demand Saturday night leisure window.
- **Scheduled downtime:** Sunday has the lowest demand and revenue potential. This period should be designated for vehicle maintenance and driver downtime, in preparation for the busy work week. This will further maximise the reliability of the service.
- **Zone prioritisation:** Drivers should be strategically positioned within high-turnover Zones. Our model identifies Zones 237, 161, 236, 230, 186 and 162 as the most reliable locations for consistent trip volume and revenue density.

7 Limitations

Although our models are highly accurate for typical operations, specific limitations must be taken into account. Our models rely on historical and temporal patterns. They do not integrate real-time sudden weather changes (e.g., heavy snow) or unscheduled special events (e.g., protests or sudden road closures), which can cause temporary spikes or drops in demand. Furthermore, since the models are trained on 2024 NYC Taxi data, this cannot account for future urban shifts. This includes potential changes in work from home policies or adjustments to public transport infrastructure. To mitigate these limitations, future iterations of the model could include real-time weather and traffic feeds, as well as a re-training pipeline that updates the model daily with internal platform data.

8 Conclusion

By leveraging 2024 NYC Taxi data, we have developed a predictive framework that accurately

simulates the Manhattan market. With 91% accuracy in demand forecasting, this model allows the Operations Manager to proactively align supply with demand, ensuring a competitive service that maximises passenger satisfaction through reduced waiting times. Furthermore, we achieved 90% accuracy in revenue forecasting, providing a robust financial baseline that ensures sustainable driver earnings. By implementing the proposed recommendations, the company can navigate the complex Manhattan market with data-driven confidence, turning high-density urban demand into a consistent operational advantage.